

Project Report: Advanced Propulsion Systems & Simulation Design

Subject: From Experimental Rim-Driven Fans to Hybrid LRE/Antigravity Simulation **Date:** October 2023 **Author:** Propulsion Design Suite (AI Assistant)

1. Executive Summary

This report details the engineering evolution of a propulsion project inspired by Integza's "levitating" jet engine. It covers the transition from a fidget-toy-inspired air-cushion fan to the rigorous design of a 500N Liquid Rocket Engine (LRE), and finally, the architecture of a dedicated software suite that integrates Gemini AI for design optimization and theoretical antigravity research.

2. Case Study: The Rim-Driven Levitating Jet

2.1 Concept Inspiration

The project began by analyzing the **Metmo fidget toy**, which utilizes a high-precision air cushion to allow a heavy metal cylinder to fall slowly and rotate with minimal friction.

2.2 Technical Failures & Solutions

- **The Thermal Seizure Paradox:** In the initial prototypes, the aluminum fan expanded at high temperatures (350°C+), exceeding the tolerance gap between the fan and the brass housing.
- **Engineering Pivot:** The creator utilized **water injection** to generate steam. This provided two benefits:
 1. **Phase Change Cooling:** The energy required to turn water into steam absorbed latent heat, preventing the aluminum from reaching its seizure point.
 2. **Pressure Augmentation:** The volume expansion of steam increased the mass flow and "levitation" pressure without requiring extra fuel.

3. Fundamental Theory: Liquid Rocket Engines (LRE)

To move from experimental toys to aerospace-grade hardware, we analyzed the thermodynamics of the **LRE-500** (a 500N thrust engine).

3.1 Governing Equations

- **Thrust (F):** $F = \dot{m}V_e + (P_e - P_0)A_e$

- **Specific Impulse (I_{sp}):** A measure of efficiency, defined as thrust per unit weight flow of propellant. For RP-1/LOX, typical sea-level I_{sp} is ~280-300s.
- **De Laval Nozzle Geometry:** The engine uses a convergent-divergent nozzle to accelerate subsonic combustion gases to supersonic velocities.

4. Design Specifications (LRE-500)

Based on a target thrust of 500N and a chamber pressure of 20 Bar, the following dimensions were established for CAD modeling:

Component	Dimension	Purpose
Chamber Radius (R_c)	25 mm	Mixing and combustion volume
Chamber Length (L_c)	80 mm	Residential time for complete combustion
Throat Radius (R_t)	11 mm	Flow choking and mass flow control
Exit Radius (R_e)	38 mm	Supersonic expansion ($\epsilon \approx 12$)
Nozzle Length	100 mm	Efficient gas acceleration

5. Software Architecture: The Design Suite

A functional software prototype was developed to automate these calculations and provide an interface for future research.

5.1 Technology Stack

- **Frontend:** React.js with Tailwind CSS for a high-fidelity "dark mode" aerospace UI.
- **Visualization:** SVG-based parametric drawing and Recharts for static fire simulation telemetry.
- **AI Core:** Integration of `gemini-2.5-flash-preview-09-2025` for:
 - **Design Optimization:** Real-time feedback on parameter changes.
 - **Metallurgy Advising:** Selecting alloys (Inconel, Niobium) based on heat flux.
 - **Text-to-Speech:** Natural language briefings using `gemini-2.5-flash-preview-tts`.

6. Future Frontier: Antigravity Research

The software was expanded to include a **Theoretical Antigravity (AG) Mode**. This template shifts the focus from Newtonian reaction mass to quantum-level field interactions.

6.1 AG Parameters

- **Graviton Flux (THz):** The density of theoretical gravitational particles.
- **Field Intensity (T):** The magnetic/electric field strength required to manipulate the local gravity well.
- **Casimir Pressure (nN):** Utilizing vacuum fluctuations for localized mass reduction.

7. Conclusion

This project demonstrates the synergy between experimental prototyping and software-assisted engineering. By bridging the gap between physical hardware (the LRE-500) and theoretical simulation (Antigravity mode), we have created a platform for the next generation of propulsion designers.

End of Report